SCHOOL OF ENERGY, GEOSCIENCE, INFRASTRUCTURE & SOCIETY

INSTITUTE OF PETROLEUM ENGINEERING
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DATA SCIENCE COURSEWORK 2024

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TITLE: ANALYSIS OF THE APPLICATION OF MACHINE LEARNING FOR EFFECTIVE INTERPRETATION AND CHARACTERISATION OF THE RESERVOIR.

Introduction

The main objective of this study was to examine the applications of machine learning, methods/techniques to efficiently classify litho-facies and predict heterogeneity patterns of petro-physical properties of the reservoir for better characterisation. The accurate categorisation of the reservoir is important for the exploration and production of oil and gas because it helps create an accurate model of the reservoir and can be able to define the type of fluid content, saturation and fluid flow behaviour of the reservoir.

The algorithm is trained using wireline log datasets combined with geological textural features such as grain size and grain pore size. The wireline data sets provide measurements of the Petro-physical properties of the reservoir, such as porosity, permeability, and bulk density which offers knowledge of the reservoir quality and fluid content, while the textural features provide knowledge on the deposition environments which could assist in the classification and prediction of the litho-facies. The combination of these datasets provides a variability of information for the machine learning experiment resulting in accurate classification and prediction of the litho-facies. The model can group the datasets into populations according to their unique characteristics and thus predict the litho-facies. These experiments enable geologists and petroleum engineers to better understand the reservoir and create 3D models of the reservoir. This helps to predict the fluid flow of the hydrocarbons and thus optimises production techniques.

The training employs two machine learning techniques: supervised learning, and unsupervised learning. Supervised learning involves the teaching of the model using data sets that are labelled according to the core interpretations of their respective facies type. The algorithm can then predict unknown rock types in other wells and unknown parts of the reservoir. Unsupervised learning is based on training the model with wireline log datasets that are unlabelled. The model groups the datasets into clusters according to the similarity in features and from this the geologist can then interpret these different clusters and predict possible litho-facies represented by each cluster.

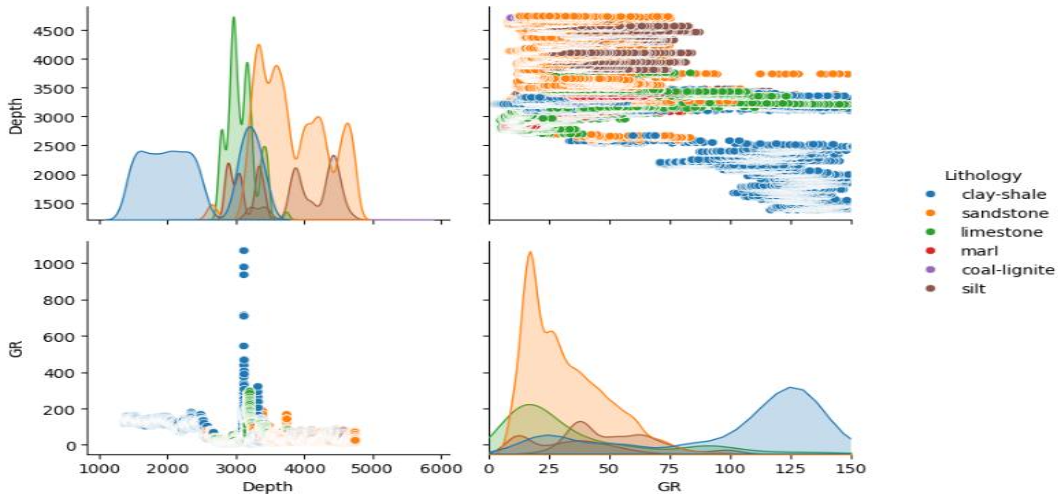
Using Unsupervised Learning (K-Means Clustering)

This algorithm groups several data sets based on the similarity of the features. The algorithm creates an average of the data sets known as centroid and groups the data into clusters according to their nearest centroid. This model simplifies complicated data for better analysis and interpretation. The algorithm redefines the centroid until all the datasets accurately fit into the cluster. The number of clusters quantifies the level of variability of the reservoir i.e., it could tell how the different petro-physical properties and wireline responses vary along the formation. A high number of clusters shows high grade heterogeneity of the reservoir, and a low number of clusters shows a low grade of heterogeneity. Knowing the diversity of the data could give a better understanding and characteristics of the reservoir. Knowing of fluid saturation and fluid flow could support in effective planning of the production methods such as where to drill and perforate in a well and deciding the best recovery methods.

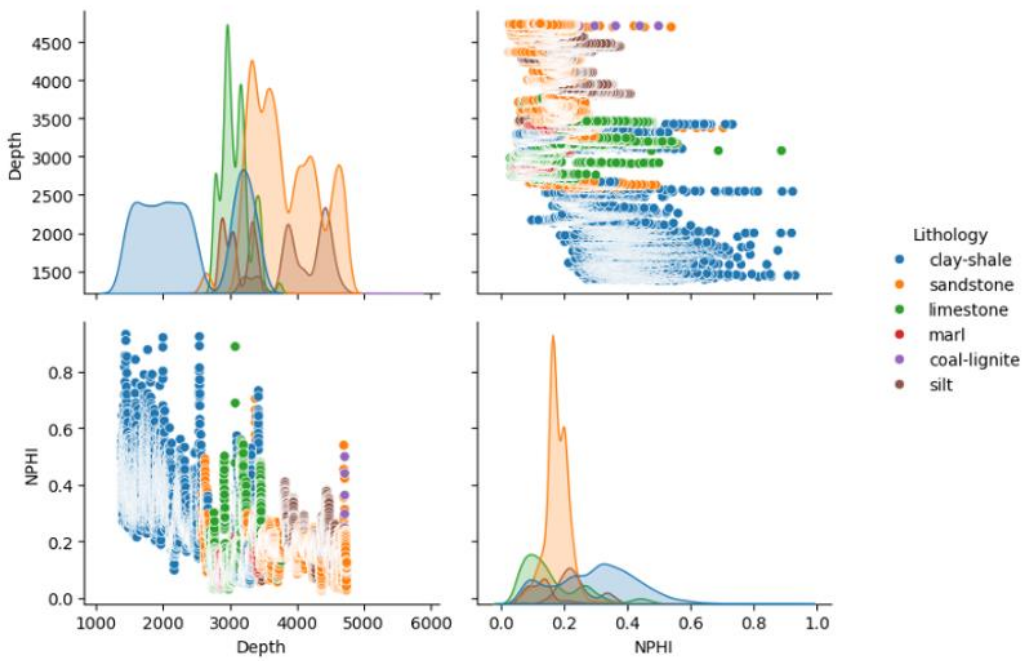
RESULTS AND DISCUSSION

Clustering wireline log data using the K-Means algorithm

Pair plot of GR vs depth of the different lithologies.



pair plot of NPHI vs Depth of different rock types



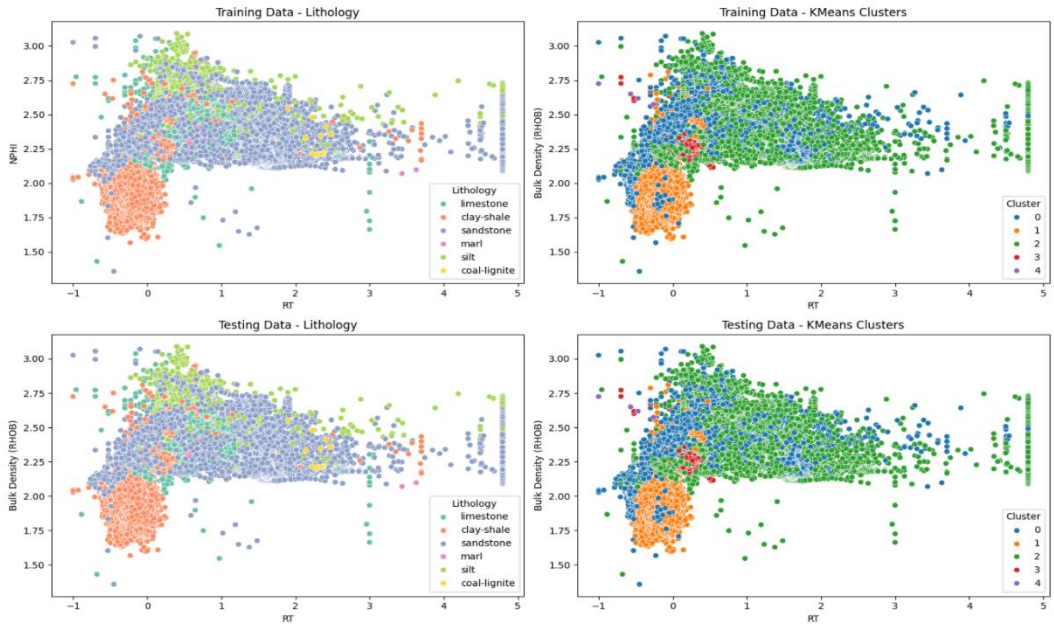
Interpretation And Discussion.

K-Means clustering algorithm was used to visualize the wireline log data sets (gamma ray and NPH values) of different lithology. The algorithm formed 7 clusters of gamma ray and NPH vales respectively with each cluster demonstrating a specific range of values of gamma ray and NPH values. The high values indicate a high clay content in formation and a cluster with high values indicate high amount of hydrogen in the reservoir. Since shale has hydrogen in bound water the hydrogen absorbs the neutron thus resulting into a high value of neutron. The analysis of the distribution of the gamma ray values and the NPHI values into clusters by the algorithm would

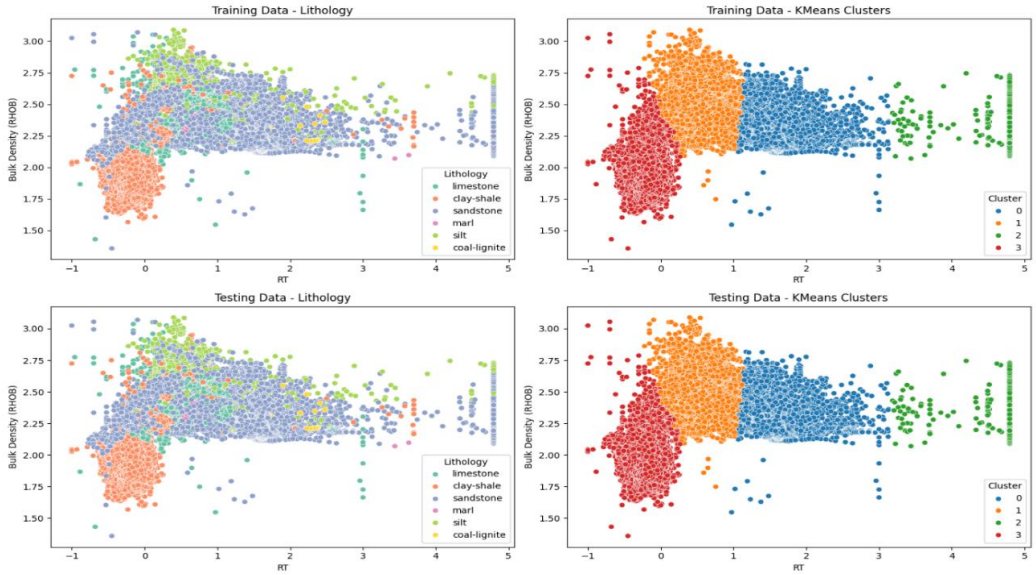
enhance the prediction of the litho-faces. A cluster with high values of gamma ray values and high values of NPHI values indicate shale at that specific depth shown in the cross plot. A cluster with low values of gamma ray values and low NPHI values could indicate sandstone. Therefore, this shows that a combination of the two types of well log data could increase the classification accuracy of the algorithm thus enabling for more understanding and characterization of the reservoir.

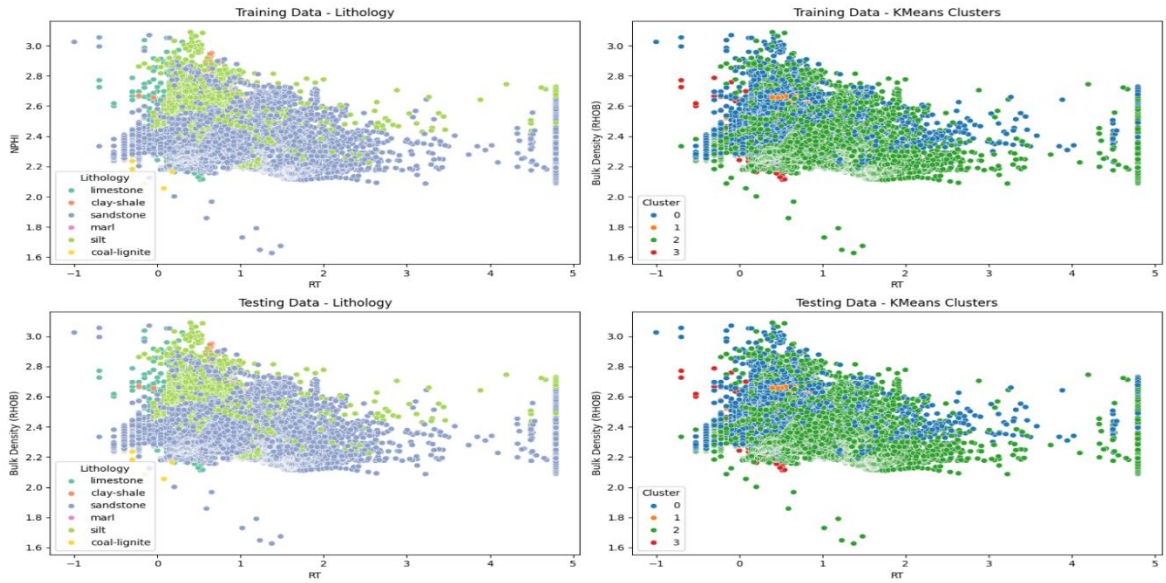
Cross Plots of Bulk Density VS Gamma Ray for Lithology Determination.

With gamma ray values



without gamma ray values.





Interpretation Discussion

The presence of gamma ray values impact on the performance of the K–means clustering algorithm compared to when data sets are visualized without gamma ray. Gamma ray values greatly rise in the presence of radioactive elements in formation. When such data is combined with values of very low gamma responses it could result in discrepancies in the ranges of data. When the algorithm creates clusters, it does so by forming centroids and assigning each value to a cluster that is nearest to the centroid. The difference in the clusters formed with training data and testing data could result from poor data pre-processing or data cleaning. if the data set for the testing wells are not scaled or cleaned as with the training data set then the model could result in inconsistences of the clusters formed thus degrading the performance of the algorithm however gamma ray is a strong indicator of the respective formation because it differentiates between clay rich faces and non-clay rich faces. Therefore, et improves on the ability of algorithm to predict a shaley formation.

Using training and testing data set without gamma ray makes it easy for the algorithm to form clusters without the noise created by the gamma ray values. it relies on the NPHI and bulk density values to form clusters and if the data is preprocessed correctly then it will effectively form clusters. However, the algorithm cannot efficiently classify the lithologies due to less variability of the data sets.

Using Supervised Learning (Classification)

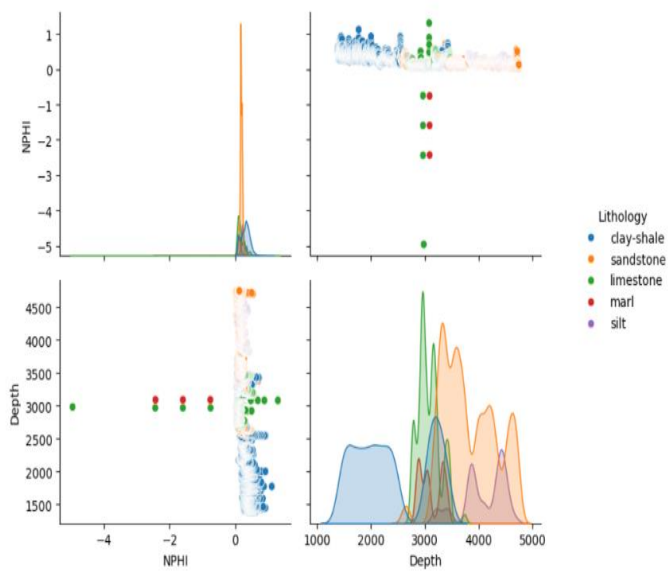
The algorithms used for the experiment are K-Nearest Neighbours (KNN) and Random Forest.
The K-nearest neighbours classify facies by identifying similar data sets and grouping them based on how close they are to each other. This closeness is determined by the metric distance from one data value to the other. Given a new k data set, the algorithm could find the closest neighbours of the data values that have the least metric distance and assign them to a class of the most common neighbouring features.

The random forest classifies the litho-facies by building random trees and combining them to make decisions and predict the specific rock type. Each tree can analyse each dataset and combine all these to form one decision of predicting the specific data set.
The two algorithms are in a position to classify the data sets through training using labelled data from interpreted litho-facies and therefore testing their knowledge to predict unknown wells.
The data is divided into three types of wells. The quality of the data that is fed into the algorithm plays a vital role in the performance of the model. That data needs to be prepared before it is fed into the model. One of the ways to prepare the data is to feature them on the same scale. Normalization of the data by placing the data sets in a similar range helps the algorithm compare the data sets easily. This is essential, especially for the KNN model that uses the distribution of the data sets to classify them. The data can then be split into Training, validation and testing wells. The training wells are used to teach the algorithm so that it is capable of detecting heterogeneity sequences such as different wireline log responses for the litho-facies and variety in the Petro-physical data possessed by the facies. Validation wells are aimed at supervising the learning process and analysing how the algorithm assesses all the different data sets given to it. This aims to prevent overfitting where the model becomes biased with the low variance data given to it and cannot accurately predict facies in new parts of the reservoir. Testing wells are used to examine how best the algorithm can classify the unknown facies and predict the specific rock type based on new data sets given to it.

CLASSIFICATION OF LITHOLOGY USING KNN AND RANDOM FOREST ALGORITHM.

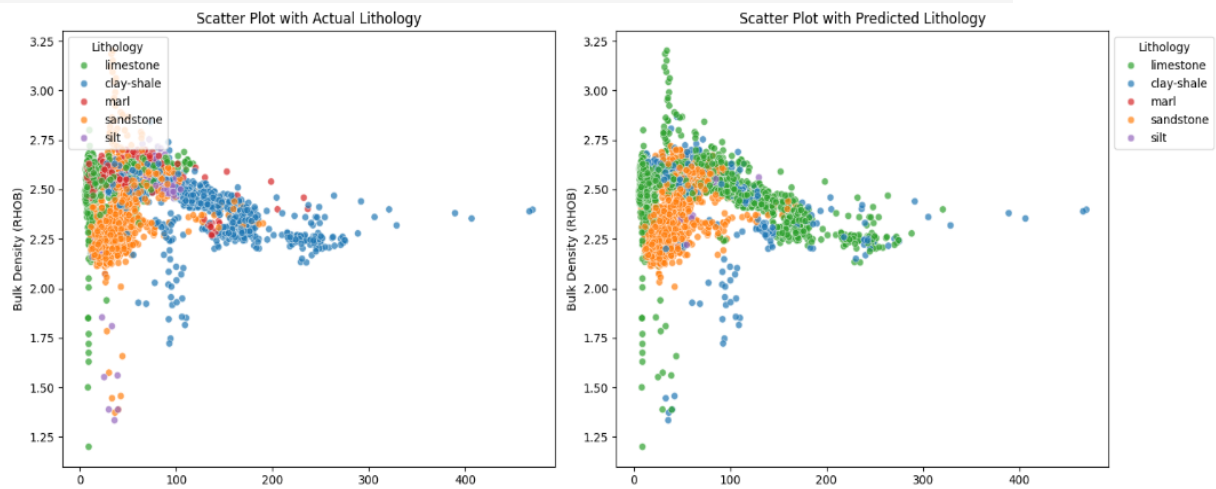
Pair plots of the wireline log data of was plotted to visualize and analyze the classification and prediction of the lithologies.

```
subset_nphi_rhob = well_data[['NPHI', 'Depth', 'Lithology']]

plt.figure(figsize=(4, 4))

# create pairplot
g = sns.pairplot(
    subset_nphi_rhob,
    hue='Lithology',
    height=3,
    diag_kind='kde',
    palette='tab10',
    plot_kws=dict(marker="o", alpha=1.0)
)

plt.suptitle('Pairplot of NPHI vs Depth (across Lithologies)', fontsize=10, y=1.02) # add main title
g._legend.set_bbox_to_anchor((1.20, 0.5))  # position legend to right
plt.setp(g._legend.get_title(), fontsize=10)  # set title for legend
plt.tight_layout() # adjust layout
plt.subplots_adjust(hspace=0.1) # adjust hspace value to increase/decrease space between rows of plots

plt.show()
```



A plot of NPHI vs depth for the different lithologies

```python
# define a color palette for the lithology classes
custom_palette = {
    'clay-shale': '#1f77b4',  # blue
    'sandstone': '#ff7f0e',   # orange
    'limestone': '#2ca02c',   # green
    'marl': '#d62728',        # red
    'silt': '#9467bd'         # purple
}

# create a figure with 1 row and 2 columns
fig, axes = plt.subplots(1, 2, figsize=(15, 6))

# first subplot: scatter plot by actual lithology
sns.scatterplot(ax=axes[0],
                x=test_df['GR'],
                y=test_df['RHOB'],
                hue=test_df['Actual_Lithology'],
                palette=custom_palette,
                alpha=0.7)
axes[0].set_title('Scatter Plot with Actual Lithology')
axes[0].set_xlabel('Gamma Ray (GR)')
axes[0].set_ylabel('Bulk Density (RHOB)')
axes[0].legend(title='Lithology', loc='upper left')

# second subplot: scatter plot by predicted lithology
sns.scatterplot(ax=axes[1],
                x=test_df['GR'],
                y=test_df['RHOB'],
                hue=test_df['Predicted_Lithology'],
                palette=custom_palette,
                alpha=0.7)
axes[1].set_title('Scatter Plot with Predicted Lithology')
axes[1].set_xlabel('Gamma Ray (GR)')
axes[1].set_ylabel('Bulk Density (RHOB)')
axes[1].legend(title='Lithology', loc='upper left')

# create a custom legend
handles, labels = axes[0].get_legend_handles_labels()
axes[1].legend(handles, labels, title='Lithology', loc='upper left', bbox_to_anchor=(1, 1))

# adjust layout for better spacing
plt.tight_layout()
plt.show()
```



Scatter plots of bulk density vs gamma-ray values of actual and predicted lithology

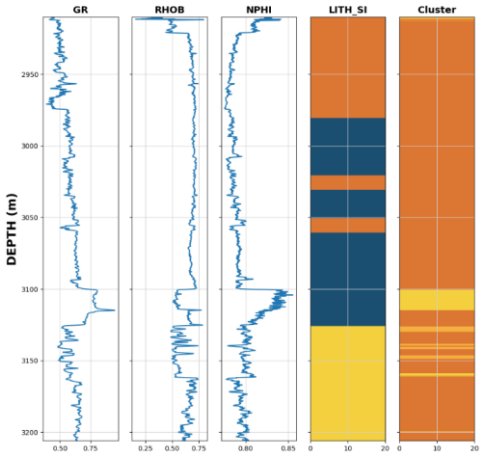
Hyperparameter optimization aids in obtaining the best combination of the data sets that would enhance the performance of the system thus giving an accurate classification of the lithologies. For the k-nearest neighbors' algorithm, the hyperparameter depends on the number of k (number of neighbors considered by each value. and the metric distance between them while for the random forest model depends on the number of decision tress formed and the type of features being used at each division. Techniques such as cross-validation are used to analyses the performance of the software. This technique involves the grouping of the data into subsets (training testing and validation sets) to support the classifier learn better and increase its ability in classification of the lithologies. The learning algorithm is taught to identify the key parameters of each rock type and

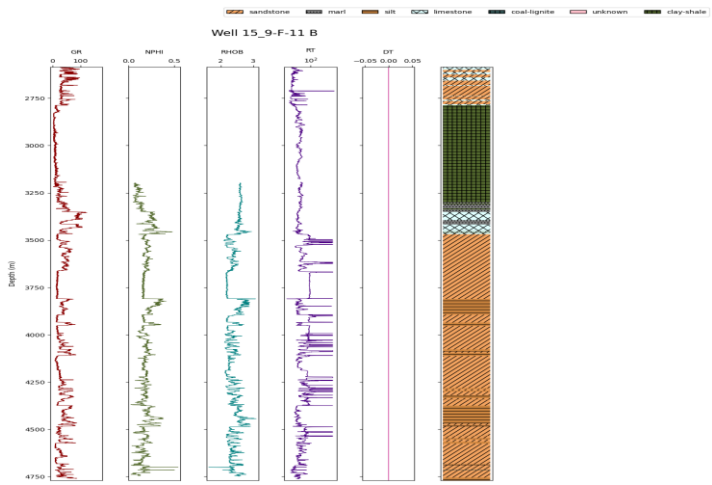
then tested with new data sets to analyze how much it has learnt. Validation can also be used to detect how much the system is learning. This technique is like a confirmatory test to see that the learning algorithm is learning correctly and is able to predict lithofacies with the given information fed to it. Therefore, selection of the data sets for the training, testing and validation is crucial for machine learning because it affects the overall optimal hyperparameter of both KNN and Random Forest algorithm. The training and testing data sets should show a various or diverse representation of the reservoir so that the classifier and learn how the petrophysical properties change across the reservoir and be able to learn to classify the different lithologies accurately without being biased. This also enables the software to predict unknown rock types with new data sets given to it.

Training set of wells chosen well 15C and 11B.

Testing set of wells chosen were well F4 and F1 B

15_9-F-15 C





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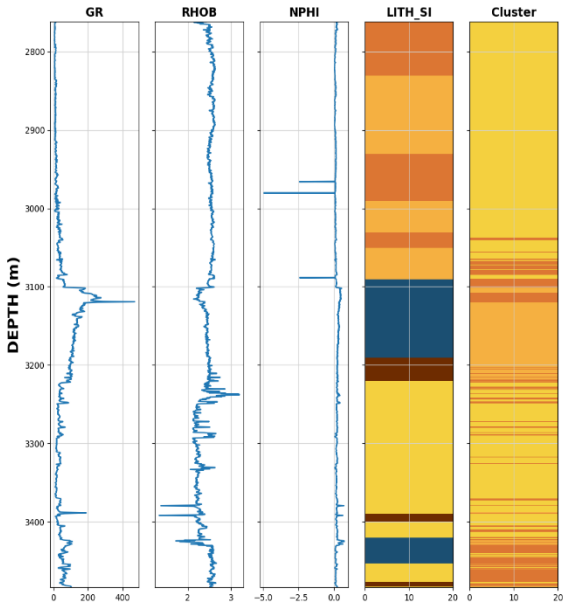
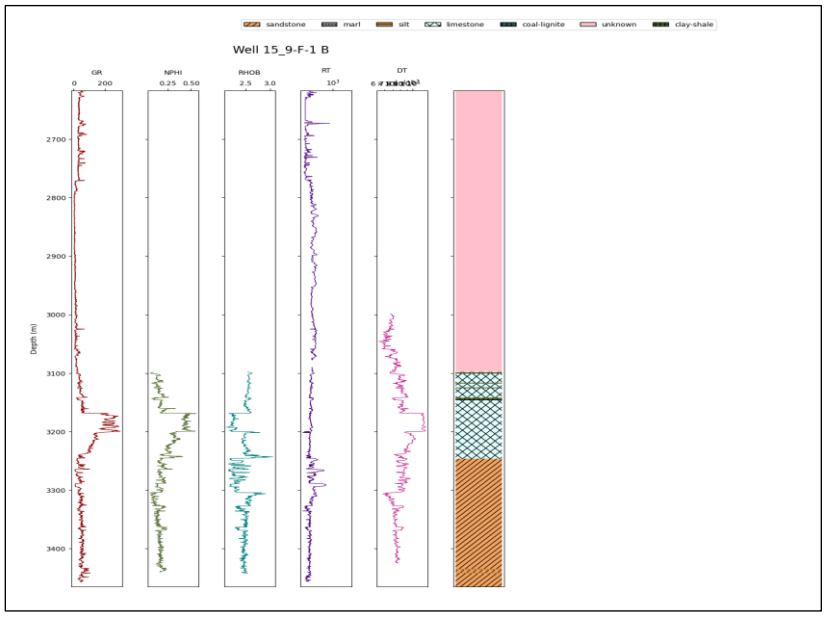
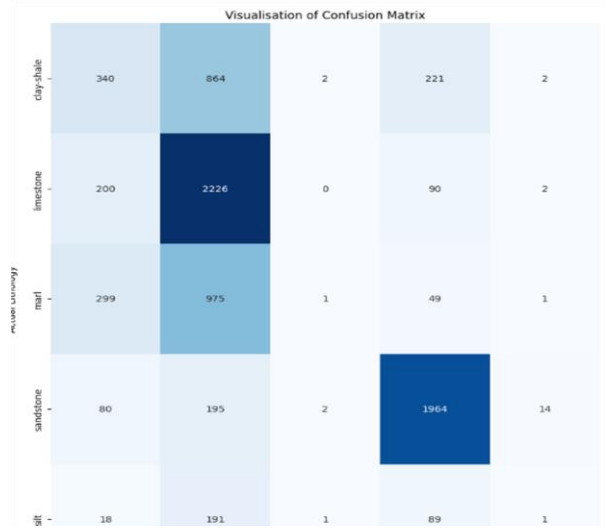




Confusion matrix of the KNN classifier.

	precision	recall	f1-score	support
clay-shale	0.36	0.24	0.29	1429
limestone	0.50	0.88	0.64	2518
marl	0.17	0.00	0.00	1325
sandstone	0.81	0.87	0.84	2255
silt	0.05	0.00	0.01	300
accuracy			0.58	7827
macro avg	0.38	0.40	0.36	7827
weighted avg	0.49	0.58	0.50	7827

Confusion Matrix: [[340 864   2 221   2]
 [200 2226   0   90   2]
 [299 975   1   49   1]
 [80 195   2 1964   14]
 [18 191   1   89   1]]
Accuracy: 58%

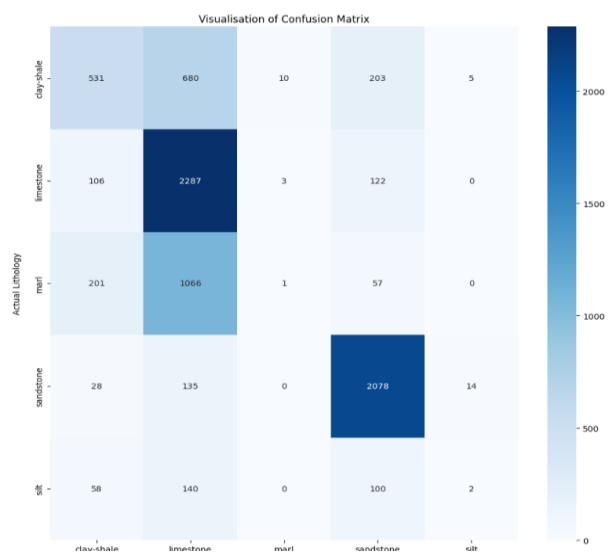


Confusion Matrix of The Random Forest Classifier

Accuracy: 0.63

	precision	recall	f1-score	support
clay-shale	0.57	0.37	0.45	1429
limestone	0.53	0.91	0.67	2518
marl	0.07	0.00	0.00	1325
sandstone	0.81	0.92	0.86	2255
silt	0.10	0.01	0.01	300
accuracy			0.63	7827
macro avg	0.42	0.44	0.40	7827
weighted avg	0.53	0.63	0.55	7827

Confusion Matrix: [[531 680   10 203   5]
 [106 2287   3 122   0]
 [201 1066   1   57   0]
 [28 135   0 2078   14]
 [58 140   0 100   2]]
Text(120.72222222222221, 0.5, 'Actual Lithology')



Comparison of the KNN and Random Forest algorithm classification results.

Both the learning algorithms were able to classify the lithologies according to their respective properties. The pair plots of NPHI vs bulk density showed the algorithms had a high ability to classify the lithologies. However, each model had its own unique advantages and disadvantages given the type of data that is input into the soft. Random forest showed a higher performance in classifying lithologies compared to KNN model, when given multifaceted datasets with a high variability. KNN is able to classify the lithologies with much simpler datasets. It was able to

classify lithologies which were clearly different from each other i.e., it was able to identify sandstone and limestone due to their difference in texture and petrophysical properties. Random forest model was able to classify and predict lithologies using intricate data sets where lithologies possess similar properties, composition and texture.

With the confusion matrix, we were able to analyze how the algorithms are performing and where the inaccuracies are occurring. Random forest had few errors compared to the KNN, in fact the random forest classifier had an accuracy of 63% and the KNN model had an accuracy of 58%. The KNN algorithm worked best when simple hyperparameters were used during the machine learning process. For example, the algorithm was able to classify sandstone with minimal error because sandstone has dissimilar properties and texture from the other lithologies so low amount of k values are required by the system. Lithologies that have almost the same grain size, porosity, permeability and sometimes wireline log response require a higher amount of k values for the learning method to be able to classify them. KNN found a challenge in being able to identify the which specific lithology possess a certain property. For the random forest, the number of decision trees made depended on the complexity of the data being input into the systems. the more sophisticated the data, the higher the number of trees were required to be able to classify the reservoir. This showed that it is essential in choosing the right hyperparameters to work with when training and testing the classifier because some hyperparameters work best with the models while others don't.

Classification accuracy of the Algorithm.

The more different the lithologies are the more the model can be able to identify the heterogeneity outline and classify the reservoir. For example, sandstone has a lower response to the gamma ray tool because it has few radioactive elements while shale has a higher response, because it has high amounts of clay content in its composition, thus making it easy for the algorithm to differentiate the two and be able to accuracy detect the litho-facies. The algorithm will have low accuracy in its performance if the lithofacies have similar properties i.e., if a single rock type was deposited in two different depositional environments such a shallow marine and deep marine then they might possess similar mineral content but different textural features especially in the grain size, making it difficult for the algorithm to correctly predict the type of lithofacies.

CONCLUSION

K-means clustering with cluster the data sets given to it with high accuracy if the data is first pre-processed, cleaned, scaled correctly. The algorithm with be able to group the specify value to the closest centroid it belongs to without coming across bodies in the data sets.

Choosing the type of training, testing and validation wells is important because it will determine how best the software will perform. When the lithologies have similar petrophysical properties then different wireline logs for porosity and information on fossil content can be introduced to visualize the changing pattern of the texture and composition of the formation, therefore increasing the accuracy of the model. Combing various datasets to represent how the reservoir is diverse in nature could aid the software predict complex heterogeneousness patterns and be able to classify unknown lithologies when new data sets are fed into the system.